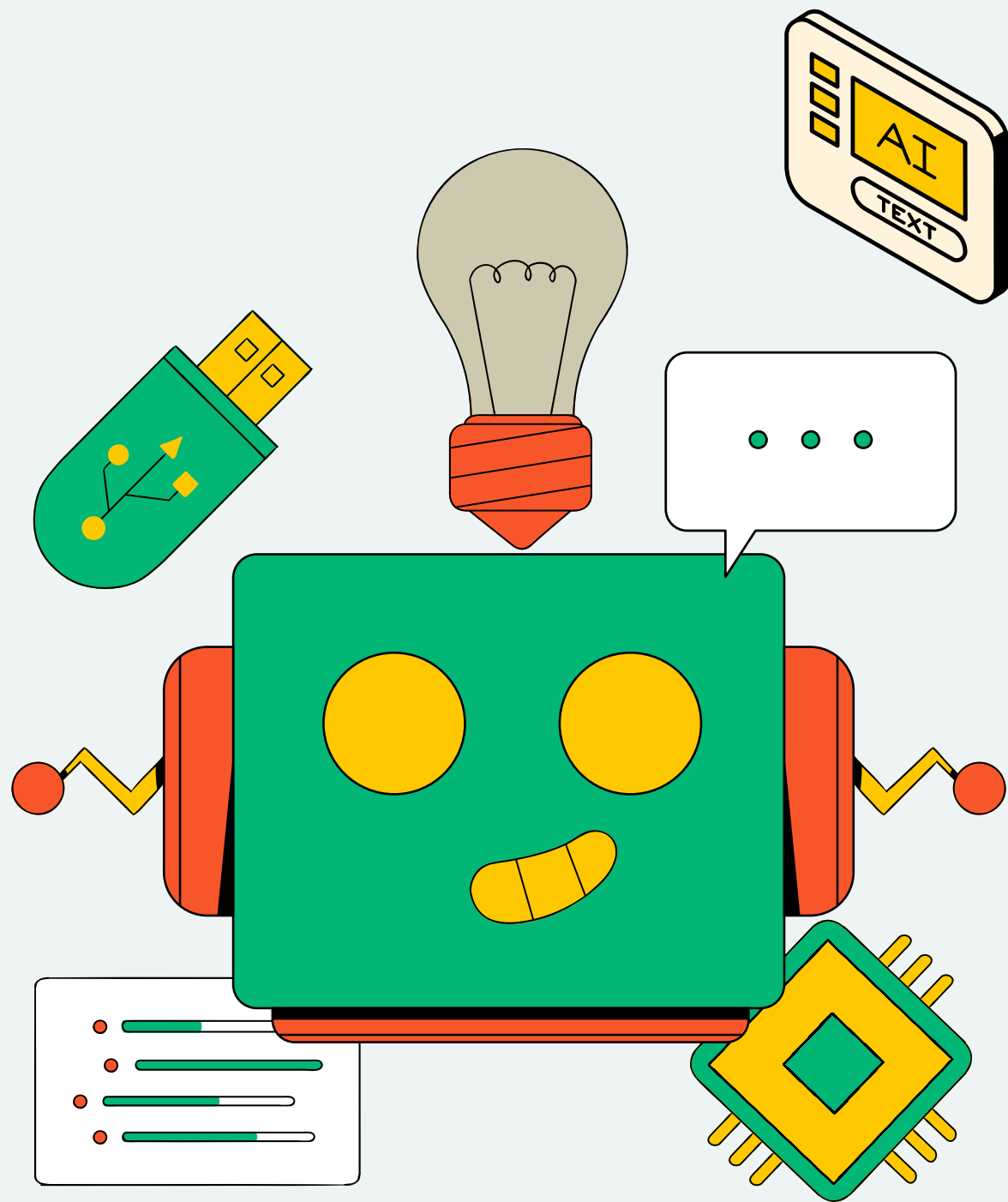


MACHINE LEARNING
5-SEM



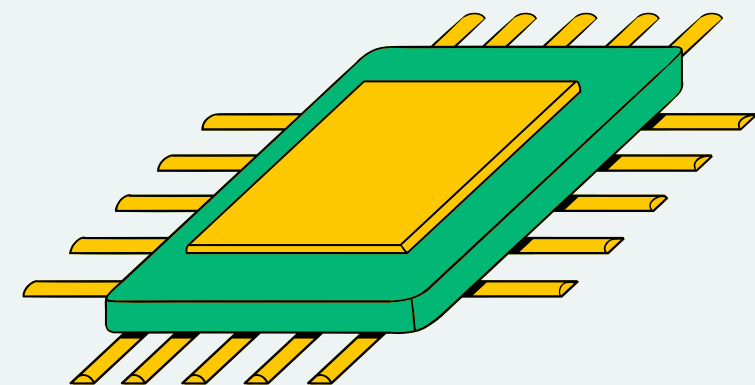
CONCEPT LEARNING TASK

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WHAT IS MACHINE LEARNING?

- Machine Learning (ML) is a subset of artificial intelligence that enables systems to learn from data and improve performance without explicit programming.
- Key paradigms: Supervised Learning, Unsupervised Learning, Reinforcement Learning.
- Focus of this presentation: Supervised learning through concept learning.
- Goal: Build models that generalize from training examples to unseen data.
- Importance: Powers applications like recommendation systems, image recognition, and predictive analytics.



INTRODUCTION TO CONCEPT LEARNING

- Concept Learning is a fundamental task in ML where the system learns to classify instances as belonging to a concept (positive) or not (negative).
- Based on inductive inference: Generalizing from specific training examples to a broader hypothesis.
- Defined in Tom Mitchell's framework: Learning a boolean-valued function from attribute-based examples.
- Training data: Instances with attributes and labels (e.g., "Enjoy Sport" example with attributes like Sky, Temp, Humidity).
- Objective: Find a hypothesis that accurately predicts labels for new instances.



THE CONCEPT LEARNING TASK

- Formal Definition: Given a set of instances X , target concept $c: X \rightarrow \{0,1\}$, and training examples D , find hypothesis $h \approx c$.
- Components:
 - Instances: Described by attributes (e.g., categorical or numerical).
 - Hypothesis: A function that maps instances to labels.
 - Training Examples: Positive (belongs to concept) and negative.
- Task: Search for the best hypothesis that fits the data with minimal errors.
- Challenges: Balancing underfitting (too general) and overfitting (too specific).



HYPOTHESIS SPACE

Hypothesis Space (H): The set of all possible hypotheses that can be formed based on attributes.

Example: For boolean attributes, H could include conjunctions (e.g., $\text{Sky}=\text{Sunny} \wedge \text{Temp}=\text{Warm}$).

Role in Learning: Efficiently search H to find hypotheses consistent with training data.



FIND-S ALGORITHM

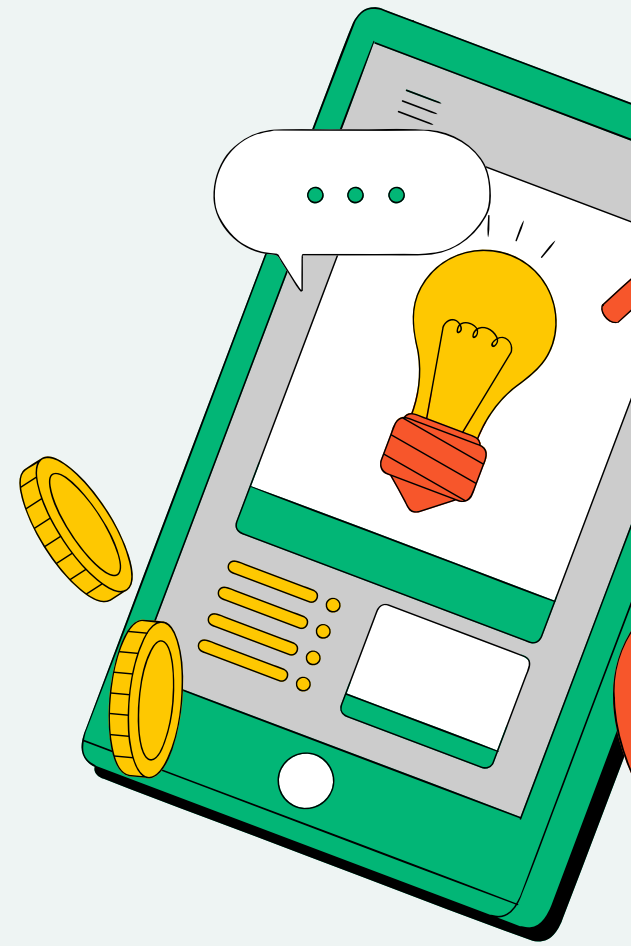
A simple algorithm for finding the most specific hypothesis consistent with positive examples.

Initialize h to the most specific hypothesis (e.g., all attributes set to null).

For each positive example, generalize h by replacing differing attributes with wildcards (?).

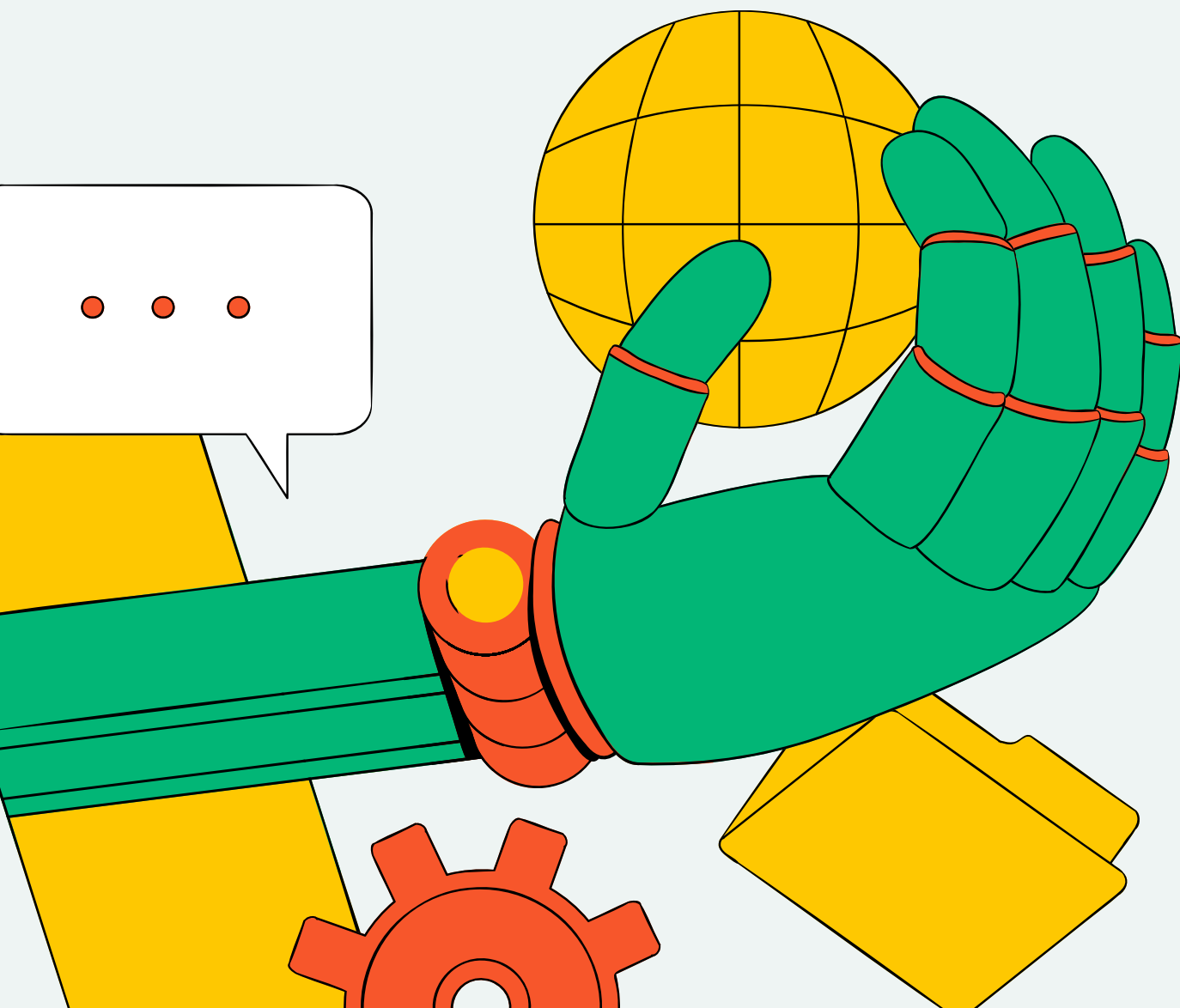
Ignore negative examples (assumes no noise).

Example: Learning "Days to play sport" – Starts specific, generalizes to Sky=Sunny, Temp=?, etc.



Steps:

CANDIDATE ELIMINATION ALGORITHM



- Candidate Elimination: Maintains a version space of hypotheses consistent with training data.
 - Version Space: Defined by Specific (S) boundary (most specific hypotheses) and General (G) boundary (most general).
 - Steps:
 - Initialize S (most specific) and G (most general).
 - For positive examples: Generalize S, specialize G.
 - For negative examples: Specialize G, generalize S.
 - Output: Hypotheses in version space for predictions.
 - Advantages: Handles both positive and negative examples; detects inconsistencies.
- Drawbacks: Computationally intensive for large hypothesis spaces.



EXAMPLES AND APPLICATIONS

Classic Example: "Enjoy Sport"
dataset – Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast;
Target: Yes/No for playing.
Find-S might yield: <Sunny, Warm, ?, Strong, ?, ?>

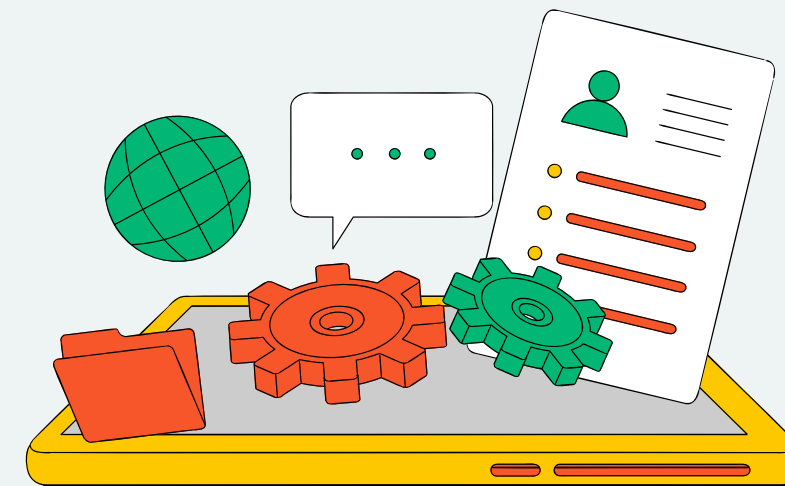
Real-World Applications:

Medical Diagnosis: Learning concepts for disease classification from patient symptoms.

Spam Detection: Identifying email patterns as spam or not.

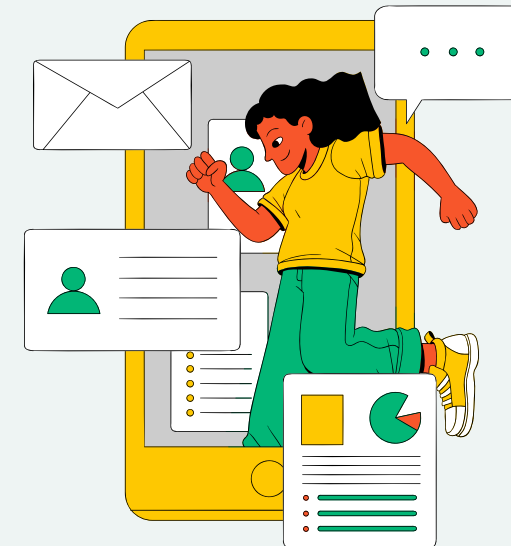
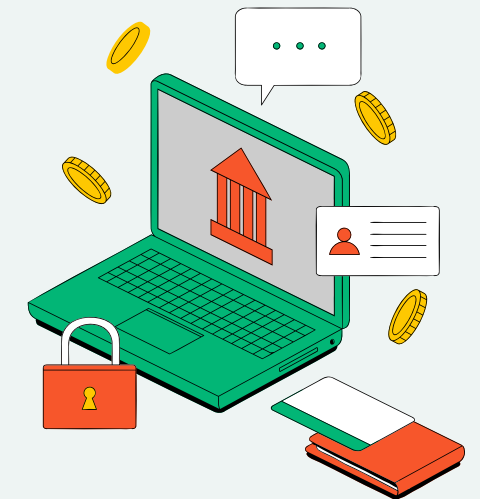
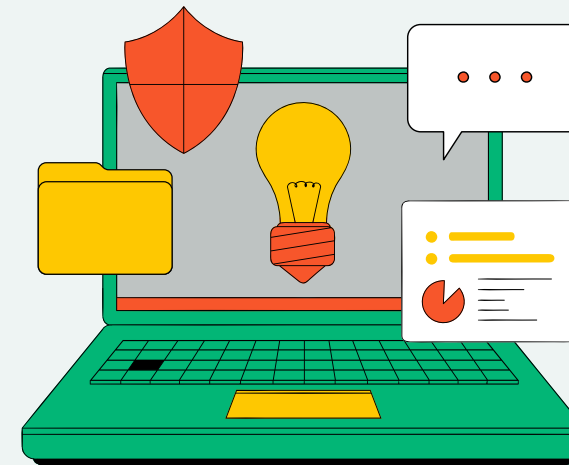
Recommendation Systems: Concept learning for user preferences.

Benefits: Foundation for more advanced ML techniques like decision trees and neural networks.



CONCLUSION

- Concept Learning is essential for understanding how ML models generalize from examples.
- Key Takeaways: Involves hypothesis search, with algorithms like Find-S and Candidate Elimination providing structured approaches.
- Future Directions: Integration with deep learning for handling complex, high-dimensional data.



A full-page background image featuring a dark, moody sky filled with heavy, textured clouds. A thin, bright crescent moon is visible in the upper center of the frame.

THANK YOU!